

KHUSHAL SINGH NEGI

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github.com/Godboyg?tab=repositories
Portfolio | Lucknow Uttar Pradesh

Summary

Full Stack Developer with hands-on experience in the MERN stack—MongoDB, Express.js, React, and Node.js. Familiar with real-time communication using WebSocket, WebRTC, Agile Development, REST APIs, MVC architecture and Git-based version control.

Technical Skills

Languages: JavaScript(ES6+), TypeScript, C/C++, Rust(Basic), HTML/CSS

Frontend: React.js, Next.js, Tailwind CSS, Redux

Backend: Node.js, Express.js, WebSockets, WebRTC, FFmpeg

Database: MongoDB, Redis, MySQL

Tools: Git, Postman, VS Code

Concepts: Data Structures & Algorithms, Networking, Operating Systems, Distributed Systems, Blockchain

Projects

Open-Chat | Live | Built a real-time chat and video communication system

Feb 2026 - Github

- Engineered an event-driven architecture enabling **sub-70ms** message latency during peak load.
- Implemented scalable socket room orchestration, optimizing fan-out broadcasts across **public + private rooms**.
- Designed distributed state-sync logic using **Redis**, enabling instant friend request + match updates across users.
- Built **WebRTC-based P2P** video calling with adaptive quality, reducing connection drops by 30%.
- Tech Stack:** Next.js, WebSockets, WebRTC, NextAuth, Elysia.js, TailwindCSS, Redis, Cloudinary, Redux.

Podforge | Live | Multi-user, recording platform with synchronized local media capture

July 2025 - Github

- Enabled 6 concurrent users with **<100ms** network latency using WebRTC and optimized signaling paths.
- Designed a local recording pipeline generating 100% lossless audio/video using MediaRecorder.
- Built a distributed FFmpeg processing workflow to merge multi-stream sessions into **HLS output, reducing export time by 40%**.
- Added live interaction controls (mute, volume, camera toggle), improving session engagement by 30%.
- Tech Stack:** React.js, Socket.IO, WebRTC, OAuth, FFmpeg, Express.js, Tailwind CSS.

White-Board | Live | A low-latency whiteboard supporting synchronized drawing across devices

September 2025 - Github

- Built socket event batching system that improved real-time draw smoothness by 25%.
- Achieved **<50ms state synchronization** with optimized diff-based updates for **7+ concurrent users**.
- Developed reusable drawing tools (pen, shapes, eraser) with resolution-aware rendering for high-FPS performance.
- Tech Stack:** Next.js, Socket.IO, Typescript, Node.js, Tailwind CSS.

Education

University of Lucknow

BCA Lucknow, India

– 2026

RLB Memorial School

Lucknow, India

Intermediate; Percentage: 64%

– 2023

High School; Percentage: 83%

– 2021

Certificate

- Samsung Innovation Campus Certificate – **SIC13505**